

LCA at Tallgrass: One-Page Brief

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How an LCA works

An LCA counts every greenhouse gas emitted and removed over a product's life. For Trailblazer, it sets each fuel's carbon intensity (CI), which scales the 45Z and clean fuel credits, and how many removal credits a tonne yields. Four phases (ISO 14040/44):

- 1. **Goal and scope:** what unit, which boundaries (corn field to Wyoming storage).
- 2. **Inventory:** every input and output, from fertilizer to booster compression power.
- 3. **Impact:** add it up in CO2e, minus land-use and supply-chain charges.
- 4. **Interpretation:** test what drives the answer (land use, grid power, leakage).

The programs Trailblazer can use

Program	Pays per	Land-use charge (iLUC)	Value	Market and outlook
45Z	Gallon, by fuel CI	Excluded after 2025	\$195–260/t (est.)	~\$12.5B/yr; ends Dec 31, 2029
45Q	Tonne stored	Not counted	\$85/t	~\$1B/yr; the floor after 2029
Clean fuel (CA, OR, WA)	Gallon below the CI target	Included in CA	\$76–147 per credit (Aug 2026)	Recovering after a 2019–25 slide; needs a CCS pathway
Removal credits (Puro, Microsoft)	Net tonne removed	Charged, about half of each tonne	\$100–200 per credit (est.)	30.4 Mt contracted in 2025; Microsoft holds 78.5%

Each tonne takes one federal credit (45Z or 45Q) plus one market credit (clean fuel or removal). iLUC is the carbon released when corn for fuel pushes farming onto new land elsewhere.

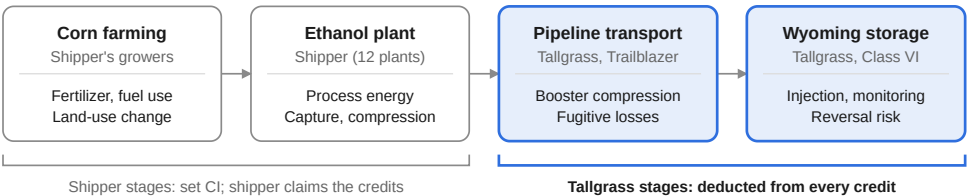
Cap-and-trade: a market to shape

Cap-and-trade is spreading. California–Quebec runs to 2045, Washington's program is linking with it, and together with RGGI these programs raise about \$6.7B a year. None of them credits CO2 stored in Wyoming today.

- **Most realistic:** push existing programs to adopt an offset or removal protocol that accepts out-of-state Class VI storage.
- **Long shot:** a Wyoming or Nebraska program. Wyoming lawmakers tried to repeal even a CCS study mandate in 2026, so a state clean fuel standard is likelier to pass.

Where Tallgrass sits

Tallgrass controls the two stages buyers audit most closely



How Tallgrass applies it

- **Own the storage-stage LCA:** one audited data package that all 12 shippers can cite, for a fee or revenue share.
- **Grade tonnes by full LCA:** premium tonnes to strict buyers (Microsoft, Frontier), the rest to spot buyers.
- **Screen new CO2 sources:** coal earns only 45Q (\$85/t) against capture costs of ~\$60–100/t, so it fits only as volume under a utility take-or-pay contract.

